Paleomagnetism of the 1.1 Ga Lackner Lake Complex and tectonics of the Kapuskasing Structural Zone[1]

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The Middle Proterozoic Lackner Lake Complex is a circular alkalic syenite – carbonatite stock with a diameter of about 5.5 km. It intrudes granulite-rank Archean gneisses in the Kapuskasing Structural Zone of the Wawa Subprovince in the Superior Province of the Canadian Shield. It adjoins the Ivanhoe Lake fault zone, which forms the boundary with the Abitibi Subprovince and is the probable locus of maximum motion between the subprovinces. Specimens from 18 sites in the complex were analyzed paleomagnetically by alternating-field and thermal demagnetization and by saturation isothermal remanence tests. Large, recent viscous remanence components required removal before a stable remanence with a mean direction of 305.4°, 64.1° (α_{95} = 5.2°) was isolated. Its pole of 53.7°N, 156.5°W (d_p = 6.7°, d_m = 8.3°) indicates emplacement at 1108 ± 10 Ma during a brief normal interval in a predominantly reversed-polarity time. This study indicates that there has been no postintrusion tilting of the Kapuskasing Structural Zone and that postintrusion uplift by unroofing did not exceed about 8 km.

Le complexe du Lackner Lake, d'âge protérozoïque moyen, est un petit massif intrusif de carbonatite – syénite alcaline ayant une forme circulaire de 5,5 km. Il pénètre les gneiss de type granulites de la zone structurale de Kapuskasing dans la sous-province de Wawa, qui fait partie de la province du lac Supérieur dans le Bouclier canadien. Il s'adjoint à la zone de failles du Ivanhoe Lake, qui forme la frontière avec la sous-province de l'Abitibi et où se situe probablement le lieu du mouvement maximum entre le deux sous-provinces. Les spécimens de 18 sites dans le complexe furent analysés paléomagnétiquement par désaimantation par champ alternatif et chauffage et par les tests de saturation rémanente isothermique. Il a été nécessaire d'effectuer une élimination des composantes importantes de rémanence visqueuse récente pour pouvoir isoler la rémanence stable ayant une direction moyenne de 305,4°, 64,1° (α_{95} = 5,2°). Son pôle de 53,7°N, 156,5°O (d_p = 6,7°, d_m = 8,3°) indique que la mise en place date de 1108 ± 10 Ma, durant un bref intervalle de polarité normale dans une période de polarité à prédominance inverse. Cette étude révèle l'absence de tout basculement postintrusif de la zone structurale de Kapuskasing et que le soulèvement postintrusif par destruction de la voûte n'excède pas 8 km.

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Introduction

The Lackner Lake Complex is a Proterozoic alkalic syenite – carbonatite pluton that intrudes Archean host rocks at the eastern edge of the Kapuskasing Structural Zone (KSZ) next to the Ivanhoe Lake cataclastic zone (ILCZ) (Fig. 1). This study was undertaken for three reasons. The first was to try to establish more precisely the time of intrusion of the complex because its radiometric age determinations span about 100 Ma. The second was to see if the complex yielded a remanence direction that matched and hence could explain known overprints in the Archean granulite gneiss and anorthosite host rocks of the KSZ. Provided the age could be constrained, then the third reason was to check for evidence of any postintrusive tectonic motion on the KSZ.

Geology

The geology of the Lackner Lake Complex has been well described by Hodder (1961), Parsons (1961), and Sage (1983). Outcrops are sparse, so most of the geological information has come from diamond drilling for niobium and apatite–magnetite deposits and from magnetometer surveys.

The complex crops out in the KSZ at the eastern edge of the Wawa Subprovince in the Superior Province of the Canadian Precambrian Shield (Fig. 1). The ILCZ passes close to the southeast edge of the complex and separates the Wawa Subprovince from the Abitibi Subprovince.

The KSZ host rocks for the Lackner Lake Complex are Archean granulite gneisses that range in composition from anorthositic–basaltic to tonalitic–granodioritic (Fig. 2). These gneisses trend northeast–southwest and dip steeply. They have been cut by the Early Proterozoic Hearst diabase dikes, which trend north–south and dip nearly vertically (Ernst and Halls 1984).

The complex is nearly circular, with a diameter of about 5.5 km. It comprises dominantly massive, coarse-grained leucocratic nepheline syenite that forms both the core and the periphery. In between are arcuate bands and partial rings of foliated malignite–ijolite that dip at 40°–65° towards the centre. Within these bands on the north side is a small zone of calcic carbonatite. On the west side of these bands is an interior zone of ijolite breccia, with abundant magnetite, and on the north side is an exterior zone of ijolite breccia called the Portage Lake Complex. The syenite and mafic alkalic rocks are cut in turn by carbonatite veins with pyrochlore that carry the niobium values, by the apatite–magnetite veins, and finally by occasional lamprophyre dikes that are not exposed. The contacts of the complex are apparently vertical, and the adjacent wall rocks are fenitized. There is no petrologic evidence of postintrusive metamorphism of the complex.

Minor late north-northeast-trending faults with displacements of a few tens of metres have been interpreted from the ground magnetometer surveys (Parsons 1961). These faults are found on the south and east sides of the complex, well away from the sampling sites and relatively close to the ILCZ (Fig. 2). There is no evidence from the circular shape, from

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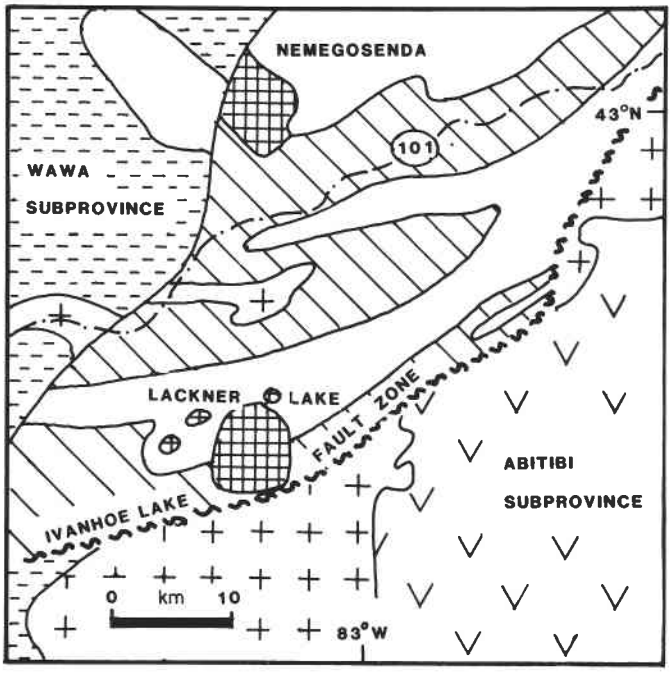


FIG. 1. Geology of the southern KSZ (after Thurston *et al.* 1975). Lithologies: Proterozoic carbonatite complexes (crossed lines); Archean granite (crosses), tonalite gneiss (dashes), metavolcanics (vees), anorthosite (clear), and granulite gneiss (diagonal lines).

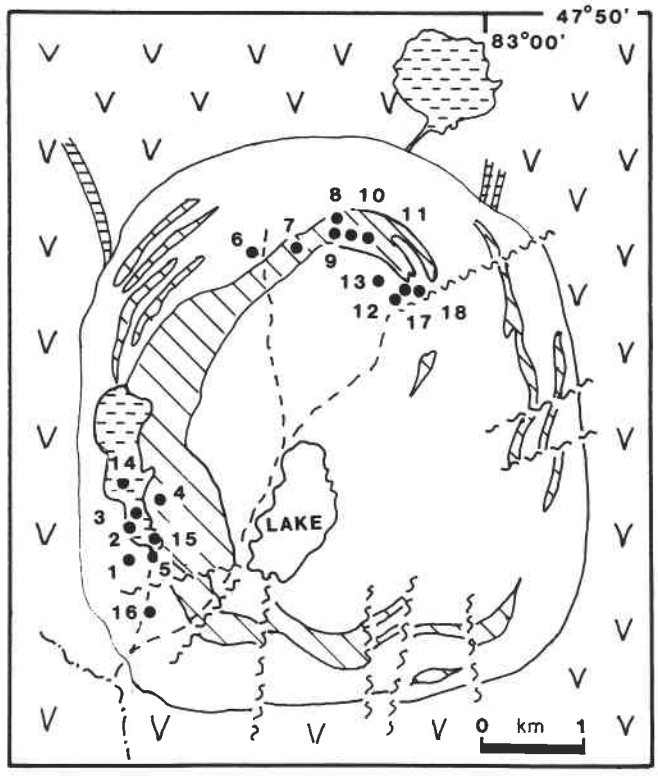


FIG. 2. Geology of the Lackner Lake Complex (after Parsons 1961) with the sampling site locations. Lithologies: Proterozoic ijolite (dashes), malignite (diagonal lines), and nepheline syenite (clear); Archean Hearst dikes (horizontal lines) and granulite and mafic gneiss (vees).

the dip of the malignite–ijolite bands, from the attitude of the contacts, or from the geometry of the airborne magnetic anomaly (ODM–GSC 1963) that the complex has been tilted. It appears to be a simple vertical stock.

The present exposure level of the complex is uncertain. Some features suggest that 10 ± 3 km might be reasonable. These features include the dominance of coarse-grained nepheline syenite, the calcic carbonatite rather than magnesian or ferroan varieties, and the sodic fenetization rather than potassic at the wall-rock contacts (Heinrich 1966; Baldcock 1973; Vartiainen and Woolley 1976). Conversely, the abundance of zenoliths and the fine-grained malignite–ijolite bands suggest that the mafic and later phases may have been emplaced in a second stage at a higher level of 4 ± 2 km (R. Sage, personal communication, 1988).

Lowden *et al.* (1963) obtained a K–Ar biotite age of 1086 ± 60 Ma using the current decay constant. Owen and Faure (1977) reported a Rb–Sr isochron of 1078 ± 7 Ma based on three whole-rock and three mineral determinations. Bell and Blenkinsop (1980) recalculated these data using the current decay constant and a 2σ error; this yielded an age of 1058 ± 22 Ma. Bell and Blenkinsop (1980) also reported a second Rb–Sr isochron of 1138 ± 29 Ma for the complex from three whole-rock determinations. When all six whole-rock determinations are pooled, they give an isochron age of 1095 ± 30 Ma (Bell *et al.* 1982).

Experimental methods and results

Sampling

Five cores were collected at each of 18 sites in the complex (Fig. 2). Eight of these sites came from the nepheline syenite, seven came from the malignite–ijolite, and three were from the ijolite breccia. Two specimens were sliced from most cores so that each site was represented by 9–14 specimens, giving a total of 185 specimens.

Natural remanent magnetization

The natural remanent magnetization (NRM) of each specimen was measured with an automated two-axis CTF cryogenic magnetometer. All measurements were done inside a shielded room with an ambient magnetic field of about 5×10^{-2} A m^{-1} to minimize the acquisition of viscous remanent magnetizations (VRM). The mean NRM intensities span a broad range from 10^{-3} to 10^{3} A m^{-1}, with the most intense values found in the magnetite-rich ijolite breccia at site 3. The mean NRM direction was found by averaging the specimen directions to get the site-mean directions and averaging these in turn to get the unit-mean direction (Fisher 1953). This mean direction of 340.6°, 72.3° (8.7°) (declination, inclination (α_{95})) encompasses the present Earth's magnetic field (PEMF) direction within its cone of 95% confidence (Fig. 3). This alignment, combined with the high NRM intensity, suggests that the NRM is dominated by modern VRM components. The removal of large VRM components in coarse magnetite to isolate the primary remanence is a common problem in syenites.

Alternating-field step demagnetization

Two specimens with contrasting NRM intensities from different cores from each site were alternating-field (AF) demagnetized in 14 steps up to 160 mT with a Sapphire Instruments SI-4 demagnetizer. The NRM intensity of almost all of the specimens decays very rapidly, giving a median destructive

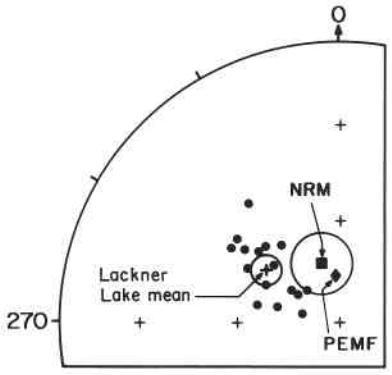


FIG. 3. Northwest segment of the lower hemisphere of an equal-area stereonet. The diamond is the present Earth's magnetic field direction. The square, with its cone of 95% confidence, is the mean NRM direction for the complex. The dots are the site-mean directions after demagnetization, and the cross, with its cone of 95% confidence, is their unit-mean direction.

field of <5 mT. On an orthogonal vector plot, these specimens show linear decay above 10–20 mT, giving a normal northwesterly direction that represents only the last few percentage of the initial NRM intensity (Fig. 4*a*). For a few specimens, the demagnetization was so rapid that no stable direction was isolated. No steep reversed directions were found. In a majority of cases, the linear segment can only be seen if one expands the scale on the orthogonal vector plots to see the last few percentages of the NRM intensity (Fig. 4*b*).

Thermal step demagnetization

Two specimens from another two cores from each site were thermally demagnetized in 12 steps up to 680°C with a Schonstedt TSD-1 demagnetizer. Again, most specimens show rapid NRM decay with a median blocking temperature of <200°C. The stable direction is first isolated between 200° and 400°C, is directed steeply downward to the northwest, and represents the last few percentages of the initial NRM intensity (Fig. 5). The Curie temperatures are in the 540–585°C range of magnetite, indicating that it is the main remanence carrier in these rocks. As before, no reversed-remanence components were found.

SIRM testing

Eight specimens with a range of AF step demagnetization and petrologic characteristics were subjected to saturation isothermal remanent magnetization (SIRM) testing. These specimens were first magnetized along one axis in continuous fields of 5, 10, 20, 40, 80, and 160 mT with the Sapphire Instruments SI-4 unit. A large electromagnet was used to saturate the specimens at 900 mT.

The SIRM acquisition curves (Fig. 6*a*) show that magnetite dominates the remanence from the rapid initial increase in the SIRM intensity. When the initial 100 mT portions of the acquisition curves (Fig. 6*b*) are examined more closely, it is seen that the remanence carriers fall dominantly in the multidomain (MD) to pseudosingle domain (PSD) range of magnetite. AF demagnetization of the SIRM (Fig. 6*c*) shows most clearly that MD magnetite is the sole significant remanence carrier, and this is consistent with the presence of large VRM components in magnetically soft domains.

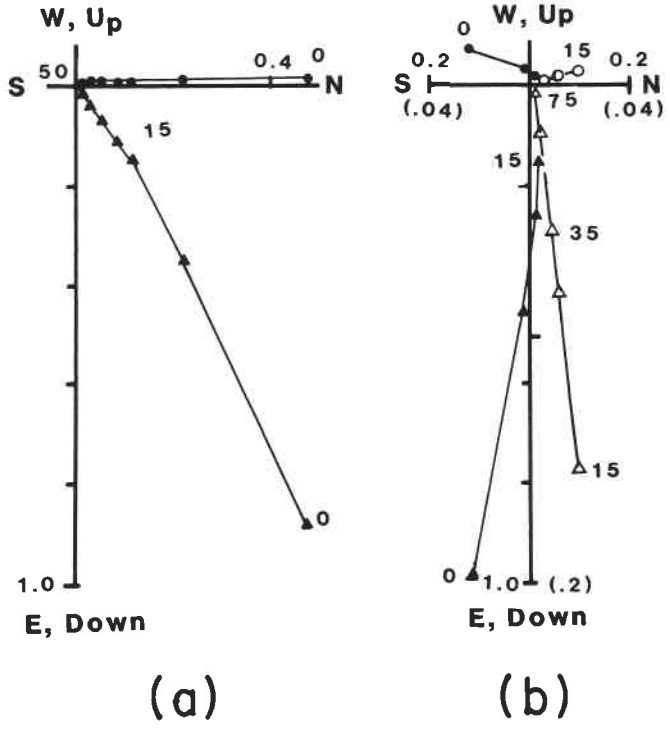


FIG. 4. Orthogonal demagnetization diagrams showing the AF step demagnetization of example specimens from (*a*) site 5 and (*b*) site 11. Horizontal- and vertical-plane projections are shown by circles and triangles, respectively. Solid and open symbols denote vectors plotted on the full and expanded scales, respectively. Axial intensities are expressed as a ratio of the NRM intensity, with the bracketed values for the expanded scale. The step intensities are in milliteslas.

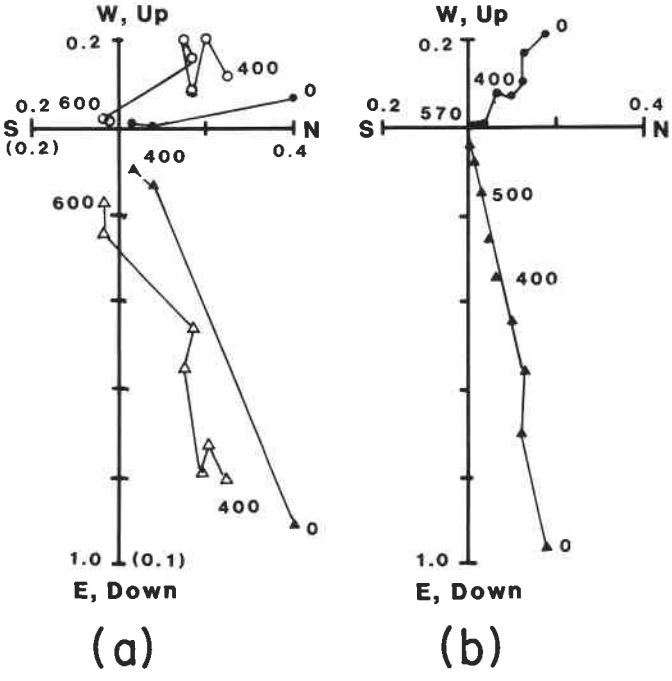


FIG. 5. Orthogonal demagnetization diagrams showing the thermal step demagnetization of example specimens from (*a*) site 10 and (*b*) site 18. The plotting conventions are the same as for Fig. 4, except that the step intensities are in degrees Celsius.

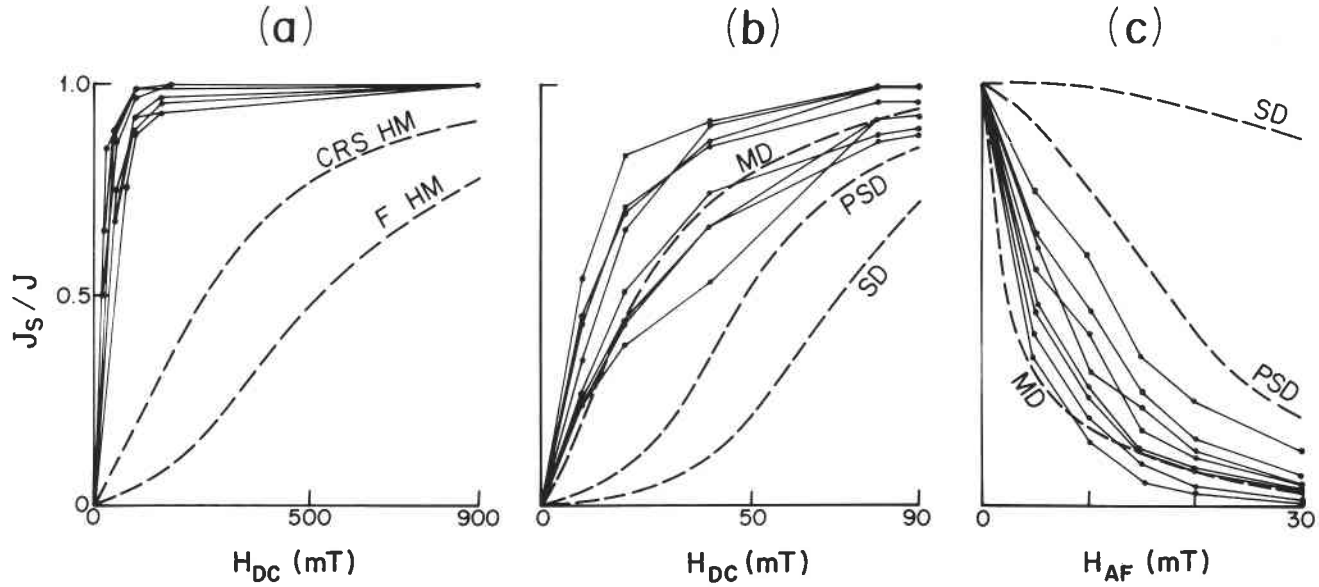


FIG. 6. Saturation isothermal remanent magnetization (SIRM) test data showing (*a*) acquisition of isothermal remanent magnetization (IRM) to 900 mT, (*b*) initial acquisition of IRM to 100 mT, and (*c*) demagnetization of SIRM. The type curves for coarse-grained hematite (CRS HM), for fine-grained hematite (F HM), and for multidomain (MD), pseudosingle domain (PSD), and single domain (SD) magnetite are from Dunlop (1971, 1972, 1973, 1981). J/J_s is the ratio of the measured to saturation remanent intensity acquired at 900 mT. H_{DC} and H_{AF} are the continuous and alternating-field peak intensities, respectively.

Final demagnetization

From the four specimens from each site that were step demagnetized in detail, it was apparent that the AF and thermal specimens were isolating the same direction with about equal precision. Therefore, the remaining six or so specimens from each site were demagnetized in at least three AF steps initially and at least one thermal step thereafter. The specific AF fields and temperatures were chosen for each site to best isolate their stable "end-point" directions. On average, this direction was reached only after 96% of the initial NRM intensity had been removed. In 23% of the specimens, it was necessary to remove >99% of the NRM to isolate the end-point. This shows that VRM forms the bulk of the NRM. One consequence is that the aeromagnetic anomaly of the Lackner Lake Complex is not distorted by its remanence. Thus, its circular shape, steep contact gradients, and lack of elongation in any direction indicate that the complex is simply an untilted vertical cylinder.

In 16 of the 18 sites, the end-point directions for 6–10 specimens formed a coherent cluster. Excluding the two sites (sites 13, 14) with random directions means that 84% of the specimens have coherent end points. The specimens that failed were mostly those with very low median destructive fields. The 16 accepted site-mean directions have radii for their cones of 95% confidence (α_{95}) that range between 8° and 15° (Table 1), and when averaged they give a unit-mean direction for the complex of 305.4°, 64.1° (5.2°) (Fig. 3) (Fisher 1953). The cone of 95% confidence for the unit mean excludes the PEMF and does not overlap with the NRM cone: it is clear that they are statistically significantly different. Also, the eight sites with the best-defined specimen remanence vectors or least VRM give statistically the same direction as those eight sites with the most VRM, i.e., 310°, 62° (8°) versus 301°, 66° (9°). These tests provide reasonable assurance that the unwanted VRM components have been successfully removed. Also, the mean directions for the nepheline syenite and later mafic phase sites are indistinguishable. Thus, either both events are essentially comagmatic and contemporaneous or the nepheline syenite was remagnetized when the later mafic phases were emplaced.

Pole position

The mean direction for the Lackner Lake Complex converts to a pole position of 53.7°N, 156.5°W (d_p = 6.7°, d_m = 8.3°). The Keweenawan or 1025–1150 Ma portion of the apparent polar wander path (APWP) for North America is called the Logan Loop (Robertson and Fahrig 1971). Recently, Halls and Pesonen (1982) and Palmer and Davis (1987) evaluated the paleomagnetic and radiometric data base for the Logan Loop, and the Lackner pole falls at about 1108 ± 10 Ma on it (Fig. 7). This paleomagnetic age agrees well with the K–Ar age of 1086 ± 60 Ma and the pooled Rb–Sr isochron age of 1092 ± 21 Ma. It is, however, a normal pole in a reversed-polarity part of the APWP. The simplest explanation is that the complex records a brief normal-polarity interval in a dominantly reversed portion of the path.

Another explanation could be that the complex is actually about 1090 Ma old, in which case its original pole position would be close to MP1 (Fig. 7). If true, then the complex would have to have been tilted about 30°W about a horizontal north-northeast axis to give its present pole. This seem improbable because the north-northeast tilt axis would be at about 45° to the east-northeast structural trends in the area and because a 30° tilt should be evident in the band attitudes, magnetic anomaly, and other geologic features. When the pole is corrected for north-northwest tilt about an east-northeast axis that parallels the ILCZ and regional petrologic trends, it moves progressively away from any known Keweenawan poles and towards the Cretaceous part of the APWP.

TABLE 1. Mean remanence statistics

Site No.	Number of specimens[a] Collected	Used	Mean remanence direction[b] Decl. (°)	Incl. (°)	α_{95} (°)	k	Demagnetization range represented[c] AF (mT)	Thermal (°C)
1	10	6	315	59	10	46	15–30	
2	10	9	322	46	13	18	15–30	450
3	10	7	299	59	12	27	15–50	560
4	10	10	303	52	10	23	22.5–50	530
5	10	10	307	55	11	21	15–60	500
6	9	8	301	75	14	17	22.5–50	560–580
7	10	8	280	65	11	27	22.5–35	560
8	9	6	282	72	11	39	22.5–130	
9	10	7	301	74	11	31	15–50	560
10	10	9	295	67	12	21	22.5–80	560
11	10	10	279	79	8	34	22.5–130	590
12	10	9	312	59	10	27	22.5–75	560
15	11	10	310	65	15	11	20–30	400–590
16	14	9	307	51	15	14	20–50	300–560
17	10	10	323	62	12	17	20–25	530–580
18	10	9	322	78	8	49	20–40	400–560
Mean		(16)	305	64	5	51	R = 15.7028	

NOTES: Sites 13 and 14 did not retain a coherent remanence.
[a]Number of specimens collected and used to compute the site mean.
[b]The mean remanence direction is given by its declination (Decl.), its inclination (Incl.), the radius of its cone of 95% confidence (α_{95}), and its precision parameter (k) (Fisher 1953).
[c]AF and thermal demagnetization treatment ranges used to isolate the specimen directions for the site.

Another possible explanation is that the modern VRM components have not been adequately removed. If correct, then the true mean direction would fall on the extension of a small circle beginning at the PEMF direction and passing through the determined unit-mean direction, giving an intermediate west-northwest and down direction. This appears to be unlikely because several sites (e.g., 1, 2, 15, 16) that were isolated by relatively low AF and thermal cleaning treatments are farther from the PEMF direction than several sites (e.g., 6, 10, 11, 17) isolated by relatively intense cleaning treatments.

Yet another possibility is that the pole reflects a postintrusive metamorphic event. This seems untenable for several reasons. First, the rocks show no evidence of postintrusive metamorphism (Sage 1983). Second, excluding perhaps the nearly contemporaneous Nemegosenda Complex, there are no known younger intrusions in the area. Third, the step-demagnetization data show no evidence of an underlying older remanence component. Fourth, the APWP is not known to pass through the Lackner Lake position at any subsequent time (Irving 1979).

Thus the best and preferred explanation is that the Lackner pole is concordant and reflects a primary untilted remanence direction.

Tectonic discussion

The main conclusion of this study is that any northwest tilting of the KSZ through listric thrust faulting on the ILCZ (Percival and Card 1983) must predate intrusion of the complex at 1108 ± 10 Ma.

The paleomagnetic data do permit some postintrusion uplift. The thermal step-demagnetization data indicate that a stable remanence component from postintrusion uplift could be preserved in magnetic domains with blocking temperatures up to about 400°C on laboratory heating. This corresponds to a geologic burial temperature of about 200°C (Pullaiah *et al.* 1975),

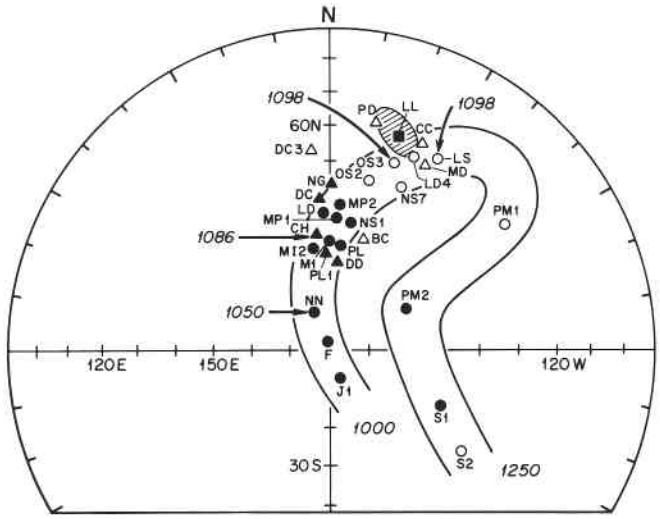


FIG. 7. Apparent polar wander path for Keweenawan time (after Halls and Pesonen 1982; Palmer and Davis 1987). The added poles are for the Lackner Lake Complex (LL, with its oval of 95% confidence, this study), the Nemegosenda Complex (NG, Symons and Garber 1974), and the Clay–Howells Complex (CH, Lewchuk and Symons 1988).

which, with paleogeothermal gradient of about 25°C km^{-1} (Blackwell 1971; Cermak and Jessop 1971), suggests the present surface of the complex could have been uplifted by as much as about 8 km since intrusion.

The Lackner Lake and 1.1 Ga Nemegosenda complexes have normal remanence directions of 305°, 64° and 305°, 46°, respectively, with concordant Keweenawan poles (Fig. 7) (Symons and Garber 1974), and are the only substantial intru-

sions emplaced into this part of the KSZ after the 2.04 Ga Kapuskasing dikes. Constanzo-Alvarez and Dunlop (1988) considered the steeply reversed east-northeast "B" remanence component found in about 25 sites of surrounding granulite gneiss and anorthosite host rocks by Symons and Vandall (1985) and Constanzo-Alvarez and Dunlop (1986) as being thermochemical overprint related to uplift reactivation of the KSZ during Keweenawan rifting, intrusion, and volcanism. Given that the normal Keweenawan complex poles are not antiparallel to the reversed "B" pole, the Constanzo-Alvarez and Dunlop (1988) interpretation seems improbable, and thus the "B" magnetization is substantially older.

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